The large amount of energy stored in the small volume of fuel makes nuclear reactors suitable as energy sources when repeated refuelling is a disadvantage. For example, a nuclear reactor was used for a long time as a power source in the Arctic. Several nuclear powered merchant ships have been built in order to determine their feasibility, but none is presently in use. The now scrapped USSR icebreaker *Lenin* (44 000 shaft horse power, launched in 1959) had three reactors of the pressurized water type, of which one was a spare. Nuclear power is used in many naval vessels in the USA, Russia, UK, China and France. The first nuclear powered vessel was the US submarine *Nautilus*, launched in 1955; in 1958 it made a famous northern voyage from the Pacific to the Atlantic, passing under the ice at the North Pole. It could travel 250 000 km (~6 times around the earth) without refuelling. At the end of the year 2000, about 250 nuclear powered vessels (8 icebreakers and the rest naval vessels - mostly submarines) were in operation. These vessels correspond to a total of about 400 reactors.

The present nuclear fleets are dominated by advanced submarines which now number several hundred. Their reactors (small PWR:s) are fuelled by enriched uranium metal (20–50% ^{235}U) alloyed with zirconium. The high initial enrichment in combination with burnable poisons and the use of extra neutron shielding to protect the pressure vessel gives a long service life of both the fuel and the reactor; from ~4 years between refuellings for the Russian icebreakers to ~30 years for the 48 MW reactors used in the French Rubis-class submarines. Hence, many submarines today have reactors that can operate without fuel replacement or pressure vessel annealing during the expected life of the submarine itself. Because of their long endurance submerged, high underwater speed, and the difficulty of detecting and destroying these submarines they are considered to act as deterrents against a nuclear war.

The world's first large nuclear powered naval vessel was the US aircraft carrier *Enterprise*, (CVN-65) which has 8 PWR reactors, producing 300 000 shp. With a displacement of ~80 000 t (and ~100 airplanes) it can cruise at 35 knots for 60 days without refuelling.

Finally, 35 small nuclear reactors have been employed by the former USSR and 1 by USA as high power energy sources in satellites. Two naval-type reactors on barges have also been designed in Russia to be used for heat and electricity supply at remote locations in the arctic. Reactors have also been developed and tested in the US for use as aircraft and rocket engines (e.g. NERVA, which was intended as a drive for a manned Mars visit).

20.2. CHEMISTRY OF WATER COOLED REACTORS

The conditions in water cooled reactors are approximately: temperatures $\leq 350^{\circ}C$, pressures ≤ 16 MPa, and intense γ- and n-irradiation. This causes potentially severe corrosive conditions.

The radiation decomposition in water cooled and/or water moderated reactors is considerable. About 2% of the total energy of the γ- and n-radiation is deposited in the water. We have seen in §8.6 that this produces H_2, O_2, and reactive radicals. In a 1 GW_{th} BWR the oxygen production is about 1 l $min^{?1}$, but it is considerably less in a PWR. Because of the explosion risk from $H_2 + O_2$, the two are recombined catalytically to H_2O in all water reactors.

Both the radicals and the O_2 formed increase the corrosion rate of reactor materials. In a BWR the corrosion rate of stainless steel (18% Cr, 8% Ni) is about 10^{-4} mm y^{-1}, leading to the release or small amounts of Fe, Cr, Co, and Ni into the cooling circuit. Many of the activation products of these are highly radioactive, e.g. ^{58}Co and ^{60}Co. The corrosion products form insoluble voluminous colloid-like products referred to as *crud* (said to be the acronym for Chalk River Unidentified Deposits). In addition to polymeric metal hydroxides, the crud contains small amounts of other materials in contact with the wet circuit: zirconium from the cladding material, possibly copper from the BWR condenser system, silicon and organic material from the water purification systems, boron from the boric acid control system for PWR, together with fuel and fission products from leaking fuel, etc. Deposition of crud on the fuel element surfaces may block cooling channels and be detrimental for the cooling of the fuel rods because of the poor thermal conductivity of crud. This can lead to a local overheating of the cladding, and a corresponding accelerated corrosion, and may cause a penetration of the cladding and a subsequent release of fuel and fission products into the reactor water. For high duty PWR-plants, local sub-nucleate boiling may pose an additional problem for high concentrations of boron in the reactor water. In this operating regime, boron accumulates in the crud, and because of the high absorption cross section for thermal neutrons in ^{10}B, an unwanted reactivity altering mechanism (parasitic absorption) is introduced. This may cause deviations in the axial power profile for individual fuel assemblies, so called Crud Induced Power Shift (CIPS), and hence affect thermal margins in other parts of the core in a negative way. In extreme cases, the axial power profile of the total core is affected, and further operation of the reactor may violate safety criteria as defined in the safety analyses of the plant. An immediate remedy, if CIPS is detected, is to decrease the power of the reactor, and replacement of fuel assemblies may be necessary during outage. These are both costly options, and to ensure that this scenario is avoided, extreme care is taken to minimise the risk for CIPS. Important parameters for optimising PWR primary water chemistry, in combination with core physics parameters, regarding the risk for CIPS are mainly pH, boron concentration and power peaking factors (ratios between local and global power).

The solubility of various corrosion products depends mainly on the temperature and pH-value, and because these properties vary throughout the primary system, different corrosion products deposit on various places in the system. In order to determine a favourable pH-value for core conditions, one should utilize the axial temperature

gradient through the core. From the core inlet, the cooling water is heated when passing through the core, and therefore, the solubility limits for corrosion products vary with core elevation. Since iron is a major corrosion product from stainless steel system surfaces, a good idea is to optimise the pH value so that iron remains in solution for the entire temperature range expected when the coolant is transported through the core, and thereby try to avoid iron depositing on the cladding surfaces (the hottest place in the core). This is a good starting strategy for minimising crud formation, and hence to prevent the problem with CIPS for PWRs.

Important corrosion products are Ni and Co, and in case of maintenance and modernisation of power plants, fresh unpassivated system surfaces are introduced (for example when steam generators are replaced in a PWR). For such cases, the source term of nickel increases immediately after introduction of high Ni-content alloys, and becomes an important source for crud formation and radioactivity distribution in the plant. A special program for "start up chemistry" may be necessary to employ after major projects, especially if power up-rates are planned in connection to modernisation of a nuclear power plant. The releases of Ni and Co are especially important to suppress, since these elements constitute the highest sources for dose rates in the plant.

Since corrosion products constantly pass the core, elements with sufficiently high neutron absorption cross sections will become activated. The radioactive corrosion products carried through the cooling loop create a serious radiation problem for the reactor personnel. Considerable effort is put into the development and selection of corrosion-resistant materials. Equally important in this respect is the selection of water conditions which minimize corrosion and the deposition of such products within the reactor core, and effective water cleaning systems. In a PWR, the crud is transported through the whole primary cooling circuit, but removed in the purification circuit. In a BWR, the crud accumulates in the reactor vessel; therefore water boiling reactors have a special cleaning circuit attached to the reactor vessel. The cleaning circuits are discussed in §§19.13.2 and 20.4. Because of the radiation problem in the plant caused by activated corrosion products, the main corrosion rates should be as low as possible, and the residence time for corrosion products in the core (activation time) should be kept to minimum by optimising the water chemistry parameters.

The amount of crud deposited in the reactor core can be partly removed during shutdown by mechanical cleaning or by washing with chemical decontamination solutions. These solutions contain mildly oxidizing agents such as alkaline permanganate and/or organic complexing agents like ammonium citrate (APAC treatment), oxalic acid or EDTA. Another way to remove crud from fuel assemblies is by ultrasonic fuel cleaning, where ultrasonic waves are used to create small oscillating air bubbles which collapse (cavitation) and case small shock waves, which in turn shatters the crud on the fuel cladding surfaces.

There are a number of ways to reduce corrosion. One is to increase the pH of the water to ~8 by adding alkali, e.g. LiOH or NH_3 ($\leq$10 ppm). While reactors of US type use ^{7}LiOH in order to reduce the formation of tritium from n,γ-capture in ^{6}Li, the VVER reactors normally use KOH. There are benefits and drawbacks from choosing either KOH or LiOH as buffer solutions, and the latter is discussed in some detail. Supposed corrosion of zirconium alloys (cladding material) for high Li concentrations has been an issue for some time. The problem arises when large amounts of boron are needed in the coolant for controlling the access reactivity at the beginning of long operating cycles on high power. In order to maintain the desired pH-value, high LiOH concentration is needed, and the corresponding risk for accumulation of Li in the cladding oxide increases. Experiments have suggested that these regimes (~10 ppm Li) may cause accelerated corrosion of the cladding, but recent findings suggest that the problem has been exaggerated. When KOH is used instead, the corrosion of the cladding is not an issue. However, activation of potassium causes an additional radiation problem in the plant (VVER).

When ammonia is used, the radiolysis yields HNO_2 and HNO_3; it is necessary to add H_2 gas to shift the equilibrium from the acidic products. At an H_2 concentration of ~2 ppm, the concentration of dissolved O_2 is greatly reduced. Instead of NH_3, hydrazine or N_2 may be added to the water to increase the pH via the reactions

$$\tfrac{1}{2}N_2 + 3H \rightarrow NH_3; \quad NH_3 + H_2O \rightarrow NH_4^+ + OH^- \tag{20.1}$$

The hydrogen content of the water must be kept fairly low to avoid brittleness of zircaloy by hydride formation. Hydride formation may also lead to mechanical deformation of zircaloy parts. Surface pre-treatment is important in order to reduce these effects.

Another measure to suppress the amount of corrosion products in the plant is to add small amounts of zink to the reactor coolant, so called Zn-chemistry. The aim for doing this is to minimise the primary corrosion of components. The principle works by substituting the chemically similar Ni (and some amounts of Co) for Zn in the surface layers of e.g. stainless steel piping and steam generator tubes. This process forms a thinner and more stable oxide surface layer which makes migration of Ni, Co and Cr through the surface more difficult. After the initial replacement of Ni in the surface matrix (passivation of the surface), the concentration of corrosion products decrease in the primary system, and a general decrease of dose rates in the plant can be achieved.

The purity of the reactor water is checked by measurement of pH, conductivity, turbidity, oxygen concentration, radioactivity, etc., either by sampling or – more usual – by on-line analyses. If the operating conditions changes, e.g. the thermal power, a chemistry transient may occur, and in such situations, the manual sampling rate of the reactor coolant should increase in order to more accurately follow any changes in the water chemistry.

20.3. BREEDER REACTORS

The possibility to breed excess fissile nuclei by consumption of fissile + fertile nuclei increases with increasing fission factor η according to §19.9. Since η depends on the fission cross section, which is highly energy dependent (Fig. 19.3), the neutron spectra in reactors must be considered. In a reactor with large amounts of highly moderating atoms, the original un-moderated spectrum is strongly shifted towards lower energies. However, even under optimal moderating conditions, a thermal Maxwell neutron distribution is never achieved in a reactor, but can be closely approached in thermal columns of research reactors. Because breeding occurs through neutron capture in fertile ^{232}Th and ^{238}U (the *only* fertile atoms in nature), with maximum probability between 600 and 700 eV, it is desirable for breeding that the neutron flux not be too low in this region. Just as breeding cannot be achieved with a thermal Maxwell neutron distribution it cannot be achieved with a fission neutron energy spectrum. Even if attempts are made to produce very "hard" neutron spectra (i.e. high energy), some moderation always occurs. Thus in a fast reactor, in which as few as possible light atoms are used, the neutron spectrum is still shifted strongly towards a lower energy region (Fig. 19.4) so that typically the neutron flux at 0.25 MeV is higher than that at 1 MeV. The curves shown for thermal and fast reactors in Figure 19.4 are only indicative, depend strongly on the particular reactor and fuel design, change with time, and vary over the core.

Because breeding is only possible for $\eta > 2$, ^{233}U is the favored fissile material for breeding in thermal reactors, and ^{239}Pu for breeding in fast reactors. Considering the shift towards lower neutron energies in fast reactors it is seen that η for ^{239}Pu is always greater than for ^{233}U. Thus ^{239}Pu can be more rapidly obtained by breeding than ^{233}U. Consequently much more effort has been put into the development of fast plutonium breeders than into thermal ^{233}U breeders.

20.3.1. The thermal breeder

In the thermal breeder reactor, ^{233}U is produced from ^{232}Th. The reactor may be designed either with a core containing a mixture of ^{232}Th and ^{233}U, or with a central zone (*core*) of ^{233}U surrounded by an outer layer (*blanket*) of ^{232}Th. However, it is necessary to minimize parasitic neutron capture in structural materials including monitoring systems, control rods, etc. Calculations have shown that a conversion ratio of 1.06 should be possible.

As an alternative to using a solid fuel matrix, both core and blanket can be liquid. In the HRE-2 (homogeneous reactor experiment), USA, the core consisted of a ^{233}U sulfate solution in D_2O, while the blanket was a suspension of ThO_2 in D_2O. The project was terminated because of various technical difficulties.